DC Generation – General Data

1. General

The DC generation system provides the aircraft with the primary source of 28 Volt DC power, and also provides the starting facility for starting the aircraft engines. (Refer to 80-00-00, Starting – General Data). The system consists of left and right starter-generator systems designed to either operate in parallel during normal operation or, if required, both systems can be operated as completely separate power systems as determined by a BUS TIE switch. A secondary source of DC power is provided by a 24 Volt main battery.

Each starter-generator system consists of a starter-generator driven by the engine auxiliary gearbox. With both generators on-line and operating as separate systems (i.e. BUS TIE switch set to OPEN) the left generator powers the left DC bus and the battery bus, and also charges the battery, and the right generator powers the right DC bus. With the BUS TIE switch set to NORMAL, the generators are connected in parallel, thus, should one generator fail, all the buses are supplied by the single on-line generator.

Generator protection from reverse current, undervoltage and overvoltage is provided by a reverse current relay, and overvoltage relay, and generator output is regulated by a voltage regulator. The overvoltage sensing is achieved by a circuit in the voltage regulator.

Generator status is annunciated on the crew alerting system (CAS) text message list. Voltage level and amperage demand are continually displayed on the electrical system window.

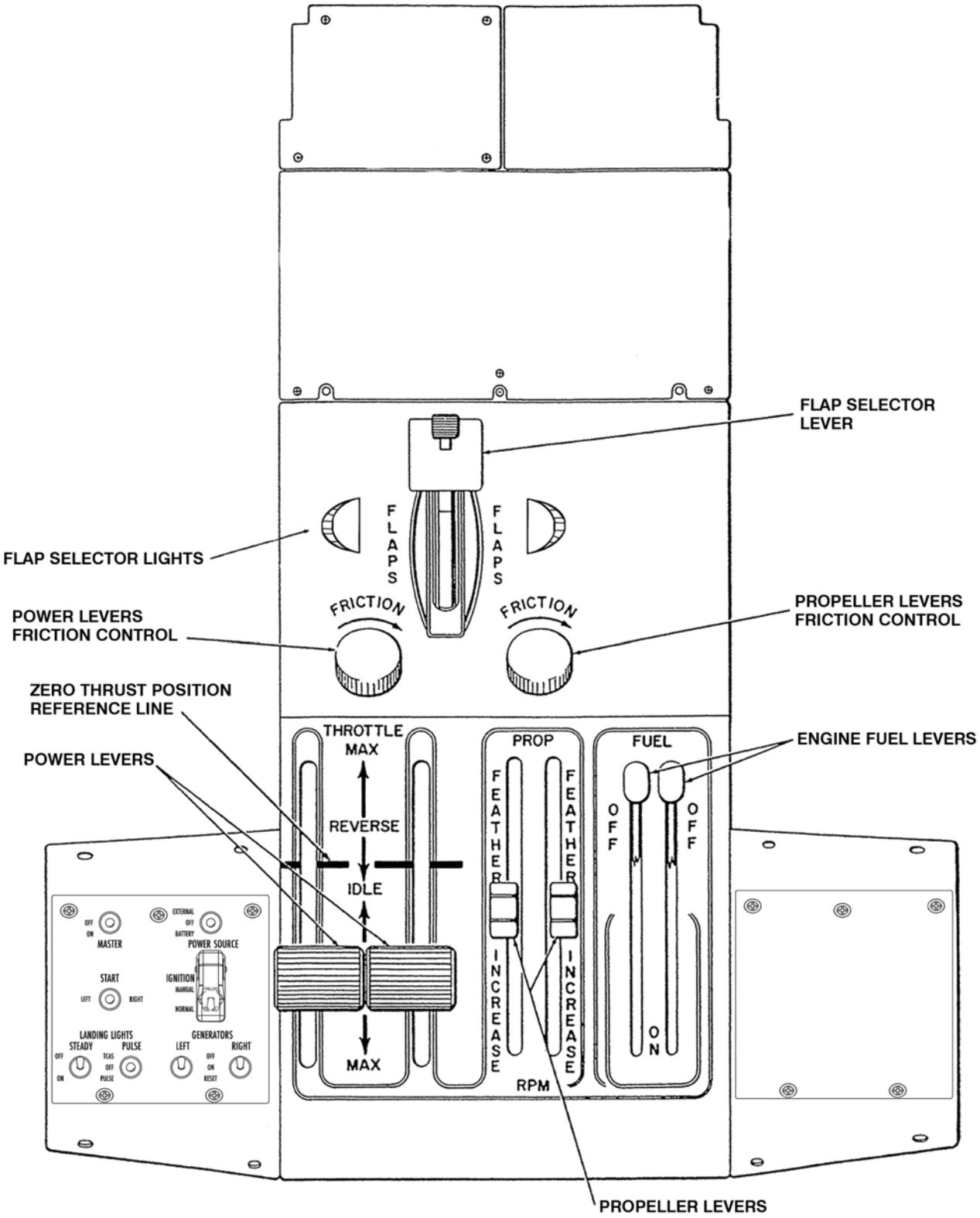
The left generator caution circuit is powered from the right DC bus and protected by a GENERATOR FAIL L circuit breaker. The right generator caution circuit is powered from the left DC bus and protected by a GENERATOR FAIL R circuit breaker. Both Circuit breakers are on the main circuit breaker panel.

Electrical system parameters are detailed in the electrical system window which is normally displayed on the lower Multi-Function Display (MFD). (Refer to 31-11-00, Display Units – General Data).

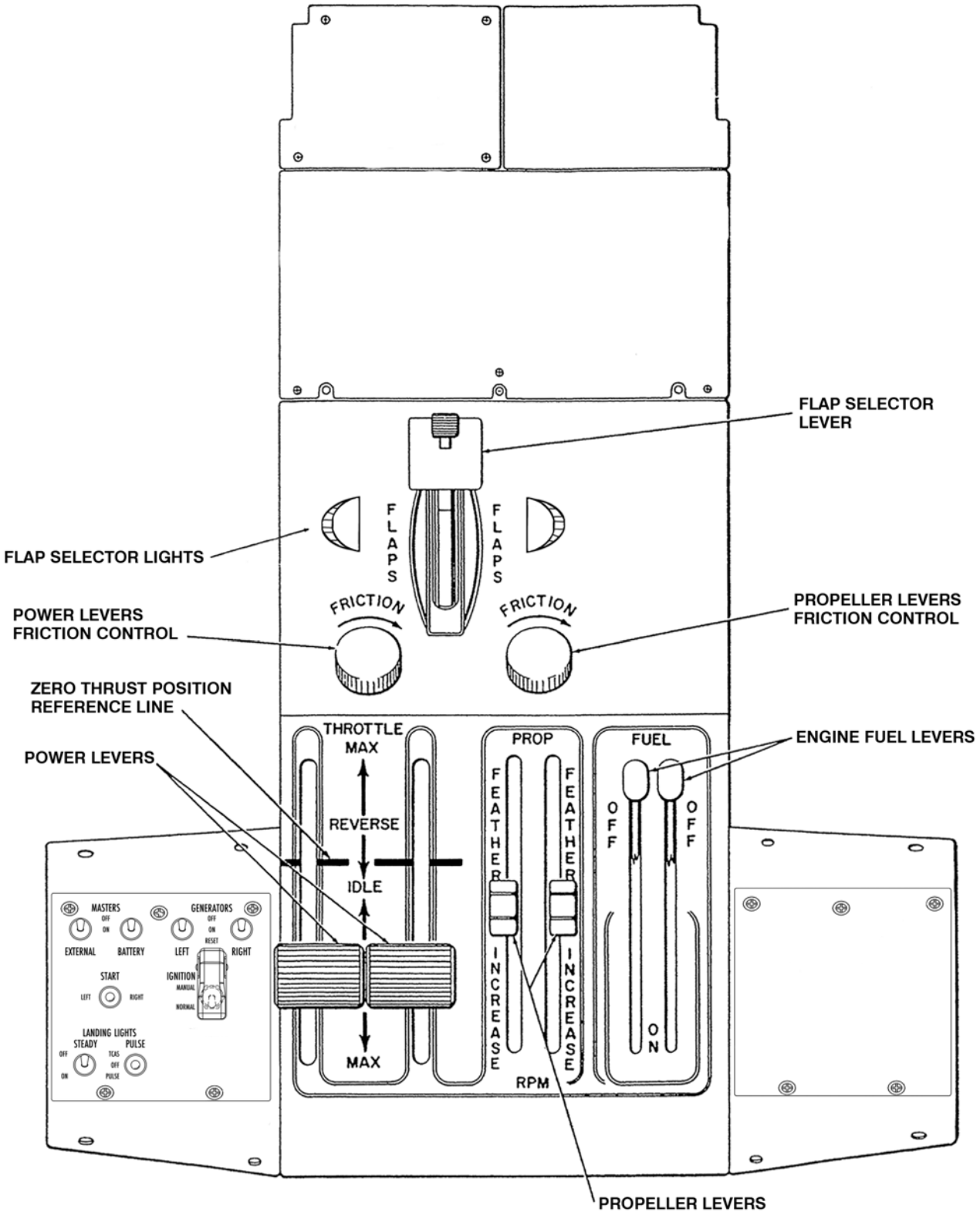
Note Mod 6/2314 (Independent External and Battery Power switches) and Mod 6/2315 (Introduction of Avionics Start Battery) affect the layout of the overhead switch panel. The differences are illustrated in Figure 1, Figure 2 and Figure 3.

Two switches labelled GENERATORS, LEFT and RIGHT, are located on the start control panel directly above the left pilot windshield (refer to Figure 1, Figure 2 or Figure 3). Each is a three position switch with positions labelled OFF, ON, and RESET. The ON position connects the applicable generator output to the electrical system through a reverse current relay and a voltage regulator. Undervoltage or overvoltage conditions de-energize the generator field relay to disconnect the generator from the system. Before reconnection can take place, the applicable generator switch must be held momentarily at RESET to restore the generator field circuit, and then released to ON.

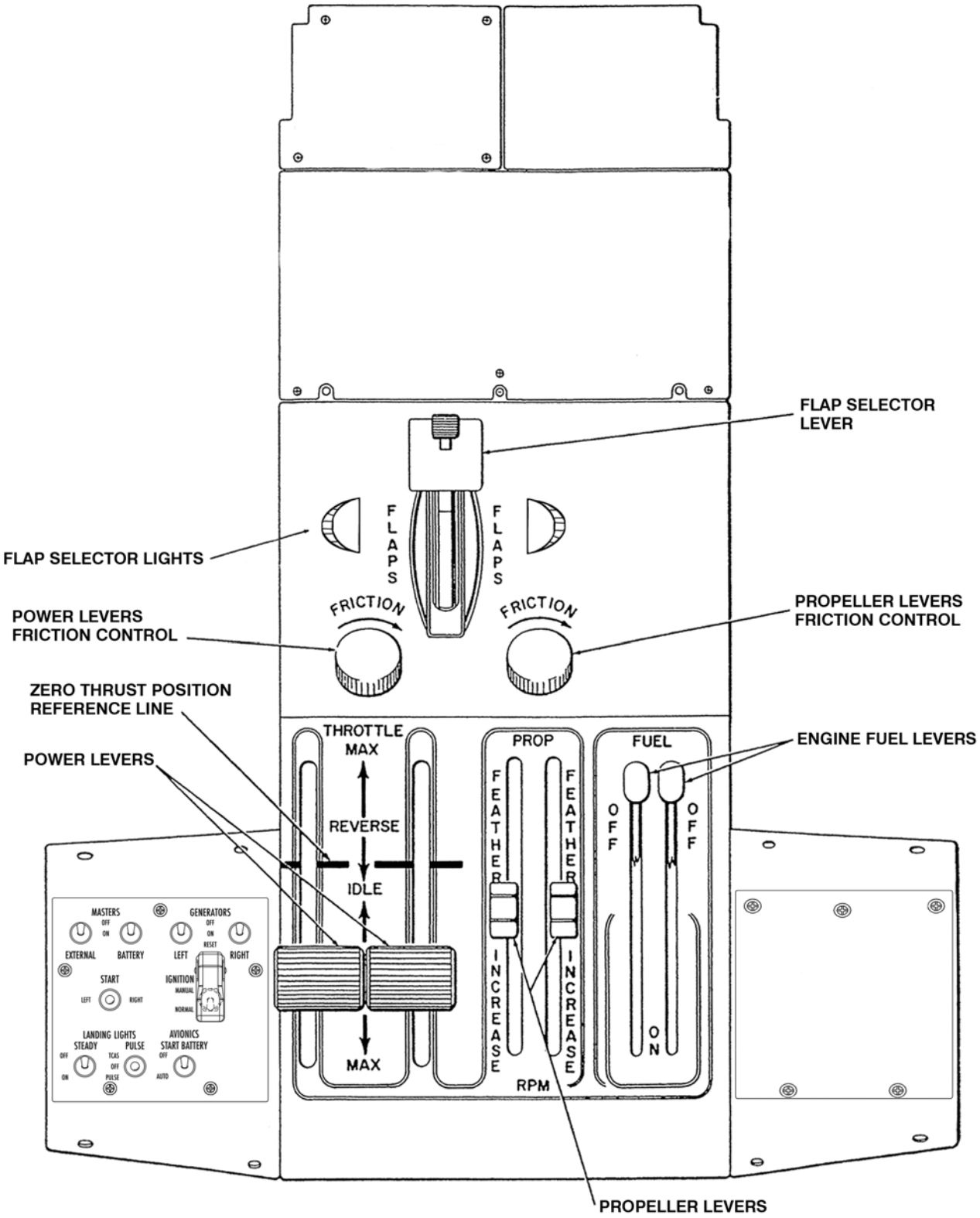
For specific details on the DC generation system refer to PSM 1-64-2W Chapter 24.



Overhead Console – Typical (Pre Mod 6/2314)
Figure 1



Overhead Console – Typical (Post Mod 6/2314) (Pre Mod 6/2315)
Figure 2



Overhead Console – Typical (Post Mod 6/2315)
Figure 3

2. Description

A. Starter-generator

A starter-generator is mounted on the accessory gearbox of each engine. These function as a direct drive starter during engine start, and as a DC generator driven by the engine once the engine is running. The starter-generators each have a nominal regulated output of 28.5 volts at 200 amperes. The full rated 200 amp capacity is available at 59% N_G and above. Following engine start a generator begins generating power when the generator switch is selected to RESET and set to ON. The field circuits are protected by GEN RESET circuit breakers, each of which is inside its respective engine nacelle top cowling.

B. Voltage Regulators

Voltage regulators in each generator circuit control the generator output at 28.5 volts nominal. At low N_G rpm (below 59%) maximum currents as depicted in the following table should not be exceeded.

N_G	Recommended Maximum Current per Generator
Idle (51%) to 57%	40 Amperes Maximum
57% to 59%	100 Amperes Maximum
Above 59% (65% recommended, refer to Note)	200 Amperes Maximum

Note Rated generator capacity is available at 59% N_G, however improved cooling is provided at higher rpm. 65% N_G is recommended for continuous loads in excess of 100 amperes.

The voltage regulators are located on the aft side of the baggage bulkhead (station 332.00). An equalizer circuit in each voltage regulator ensures equal loading (within 20 amperes) when the generators are operating in parallel (relays K5 and K6 energized, and the bus tie switch at NORMAL). The equalizer circuit is disconnected during engine starting (only relays K7 or K8 is energized). When both relays K7 and K8 are energized (IGNITION switch set to MANUAL) the equalizer circuit is connected.

The voltage regulators have an overvoltage sensing function to provide protection in the event of an overvoltage condition developing in the system during operation. An external adjustment screw and locknut is incorporated.

C. Reverse Current Relays

Reverse current relays, K1 and K2, connect starting power to the left and right starter-generators, and also connect generator output to the DC buses and batteries. The relays are mounted in the power distribution and generator control box in the RH cabin roof. Each relay unit comprises three relays. Relay 1 is energized (22 volts minimum pull-in and 18 volts drop-out) through terminal marked switch (SW). Relay 1 closes relay 2, which closes relay 3. Relay 3 senses differential voltage (0.35 to 0.65 volt for pull-in), and reverse current (9 to 25 amperes for drop-out) between terminals marked generator (GEN) and battery (BATT). When a starting circuit is applied to terminal marked APP, relays 1 and 2 are bypassed, and relay 3 is directly energized, connecting DC power from the BATT terminal to the GEN terminal for starting power input to the starter-generator. When the starting circuit is disconnected, the generator circuit is applied to the SW terminal, and normal relay operation becomes effective for generator power output through the GEN terminal, and relay 3, to the BATT terminal. With relay 3 energized, an output circuit on the terminal marked indicating (IND) is used for the generator fail relay circuit (PD-K5 or PD-K6).

In the event of an overvoltage condition developing, these relays are energized by the voltage regulator overvoltage sensing circuit to de-energize the associated generator field relay. Refer to PSM 1-64-2W.

D. Bus Feed Circuit Breakers

Circuit breakers are installed in the power distribution and generator control box in the cabin roof to protect the generation and distribution systems connected to the left and right DC buses. Additional circuit breakers are installed on the main and the avionics power distribution panels in the flight compartment for triplicate left and right DC bus feeds to the appropriate circuit breaker panels.

E. Main Battery

The main battery is a 24 volt, 48 ampere-hour, recombinant gas lead acid type, located below the floor in the rear baggage compartment. On aircraft incorporating S.O.O. 6371, the main battery has been relocated to the nose baggage compartment. The battery is provided with a cover and sealing gasket. The battery does not require provisions for overboard venting and drains.

F. Diodes

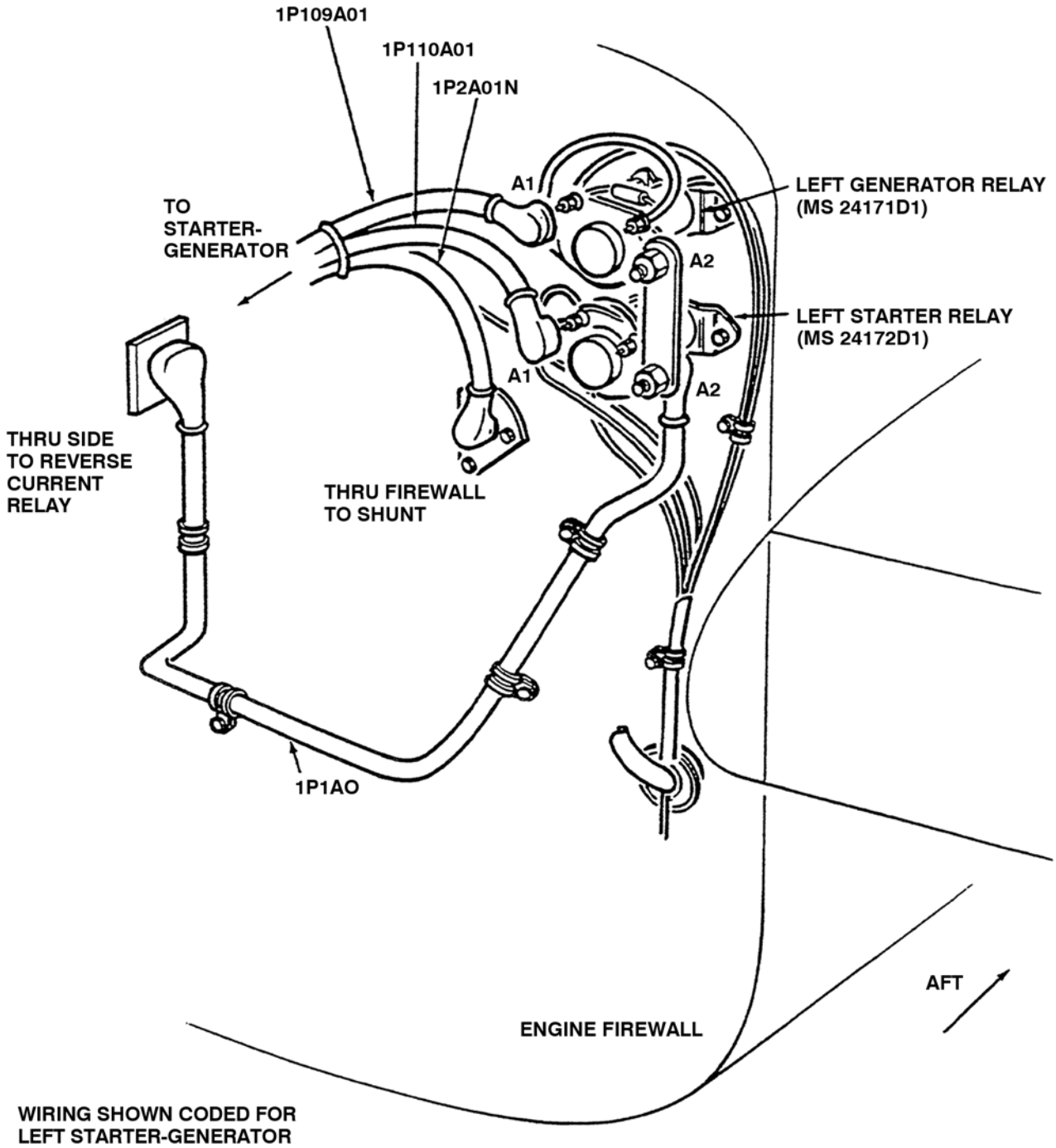
Diodes are incorporated for the following purposes:

1) To block a DC circuit feedback, installed in series with the load, in a forward biased position (anode towards the power source).

2) For DC voltage spike suppression (when a solenoid or relay coil is de-energized), installed in parallel with the coil, in a reverse biased position (cathode towards the power source).

G. Generator and Starter Relays

Left and right generator and starter relays are located on the forward face of each engine firewall (refer to Figure 4). The generator relay (MS24171–D1) is a SPST, 200 ampere continuous duty relay, the starter relay (MS24172D1) is a SPST 200 ampere intermittent duty relay. The relays are not interchangeable.



Location of Generator and Starter Relays
Figure 4

3. Operation

CAUTION **DO NOT CONNECT DC GENERATOR UNTIL ENGINE STABILIZES AT IDLE RPM (N_G) + 15% WITH BOTH ENGINES OPERATING, CONNECT GENERATORS SEPARATELY TO AVOID TRIPPING REVERSE CURRENT CIRCUIT BREAKER. AFTER A BATTERY START MAINTAIN POWER LEVER ABOVE IDLE RPM (N_G) + 15% UNTIL THE APPLICABLE GENERATOR AMPS DECLINE BELOW 100 AMPS. SWITCH OFF OPERATING GENERATOR BEFORE BATTERY START OF SECOND ENGINE.**

Refer to PSM 1-64-2W for the DC Electrical System Schematics.

Refer to Figure 1, Figure 2 and Figure 3 for details.

With the engine started (refer to 80-00-00, Starting – General Data), the engine speed stabilized at IDLE rpm (N_G) + 15%, and the starter switch released, the relevant GENERATORS control switch is momentarily selected to RESET (to excite the generator field) and then released to the ON position. Output from the generator energizes the generator relay, and through the GENERATOR CONTROL 10 ampere circuit breakers in series with a 15 ampere breaker and the GENERATORS control switch, it also energizes the generator field relay. The reverse current main relay coil is energized (via the GEN RESET circuit breaker, the GENERATORS control switch, the MASTER switch (Pre Mod 6/2314) or GENERATOR switch (Post Mod 6/2314), the generator field relay and terminal SW of the reverse current relay) to connect the generator output to the DC buses (BUS TIE switch set to NORMAL) providing the generator voltage is higher than the bus volts. The generator field relay de-energizes the external power relay (if external power is selected) to disconnect the external power circuit. When the generator is connected on-line (reverse current main relay contacts closed), the generator fail relay is energized (via the IND terminal of the reverse current relay) to disconnect the GENERATOR FAIL circuit breakers. A circuit from the left DC bus through the gen excitation breaker is connected through a resistor to terminals A of the generator and voltage regulator to maintain generator buildup and polarity of the field.

With the POWER SOURCE switch selected to BATTERY (Pre Mod 6/2314) or BATTERY switch to ON (Post Mod 6/2314), the main battery relay is energized connecting the main battery to the DC buses. With the generators on-line, the battery charge is maintained by the generators. With the generators off-line, the batteries or external power (refer to 12-00-00, Servicing – General Data), supply all the DC buses providing the BUS TIE switch is selected to NORMAL.

In the event of a generator going off-line, due to a generator overvoltage or line failure, selecting the appropriate GENERATORS control switch momentarily to RESET applies direct generator output through the GEN RESET circuit breaker to the generator field allowing the generator to build up. Moving (or setting) the control switch to ON energizes the reverse current relay to reconnect the generator on-line.

The appropriate GENERATORS control switch must first be selected OFF, to de-energize the latched generator field control relay, before selecting RESET to energize the generator field. Thereafter, releasing the control switch to ON energizes the reverse current relay as above.

CAUTION **LIMIT THE RESET OF A FAILED GENERATOR TO TWO ATTEMPTS ONLY. IF THE ASSOCIATED GENERATOR OFF MESSAGE IS NOT PRESENT BEFORE AND AFTER AN ENGINE START SELECTION (OR AFTER GENERATOR SWITCH-OFF PRIOR TO ENGINE SHUTDOWN), SET BUS TIE SWITCH TO OPEN. ALSO SET BATTERY/EXTERNAL SWITCH TO OFF WHEN LEFT GENERATOR OFF MESSAGE IS INOPERATIVE.**

4. Troubleshooting

Table 1 — DC Generation – Troubleshooting has been compiled to assist operators in the recognition of the causes, with the subsequent rectification action required, of the more commonly reported snags in the Twin Otter's main electrical system. It is hoped that this will help to prevent the unnecessary removal for investigation and repair of items which are actually serviceable or which require only minor adjustment.

It must be appreciated that the following table does not represent a complete troubleshooting guide but reflects the snags which experience has shown to be those most likely to cause the symptoms discussed.

It is assumed that the following table will be used in conjunction with the circuit schematics and diagrams.

Table 1 DC Generation – Troubleshooting

SYMPTOM	POSSIBLE CAUSE	CHECK THAT	REMEDIAL ACTION
1. (a) Low main battery output.	Poor connections of main battery ground line.	Connections of ground line to shunt and shunt to airframe are secure.	Torque nuts to 40 – 45 pound-inches. Ensure shunt contact surfaces are clean.
(b) No main battery relay output.	Open circuit diode in main battery relay control circuit	Power available at terminal A1 of main battery relay.	Replace defective diode.
2. (a) Starter will not operate when start switch selected.		Ignition switch is in “NORMAL” position.	Set correctly.
(b)	Defective relay K16 (left engine) or K17 (right engine).	No power available at terminal “B1” of relay.	Replace defective relay.
(c)	Open circuit diode CR5 – (left engine) CR6 – (right engine).	No power available at “APP” terminal of reverse current relay but power available at terminal “B1” of either K16 or K17 relay.	Replace defective diode. Check diode mounting holes are properly deburred and mica washers are undamaged and not providing short to ground.
(d)	Reverse current relay fails to close.	Power available at “APP” terminal but not at “GEN” terminal of reverse current relay.	Replace reverse current relay.
(e)	Defective starter relay.	Power available at “GEN” terminal of reverse current relay.	Check functioning of starter relay. Replace if confirmed defective.
(f)	Defective relay. K9 – (left engine) K10 – (right engine).	With relay energized terminals A2 and A3 are open from ground.	Replace defective relay.
3. (a) Generator will not come on line when GENERATORS switch is pressed to “RESET”, then set to “ON”.	Defective switch.	Continuity of switch satisfactory.	Replace switch if defective.
(b) Generator will not come on line when GENERATORS switch is selected “OFF”, then to “RESET”, then set to “ON” (Post Mod 6/1590).	Generator field control relay stuck in latched position, or normally closed contacts defective. K1A – (Left generator), K2A – (right generator).	There is continuity between terminals A2 and A3 of associated relay de-energized.	Replace defective relay.
(c) (Pre and Post Mod 6/1590 or 6/1636).	Open circuit diode CR3 (left engine) CR4 (right engine).	Power is at diode CR3 (left engine) CR4 (right engine) and continuity across diode.	Replace diode if defective.

Table 1 DC Generation – Troubleshooting (continued)

SYMPTOM	POSSIBLE CAUSE	CHECK THAT	REMEDIAL ACTION
(d)	Defective relay K9 (left engine) K10 (right engine).	With relay de-energized contacts A2 and A3 read to ground.	Replace defective relay.
(e)	Inoperative field exciter circuit.	Check voltmeter circuit breaker is made. Check continuity across resistors R1 (left engine) or R3 (right engine).	Replace defective resistor (100 ohm) if necessary.
4. (a) Generator will not stay on line.	Incorrect setting of voltage regulator, permitting overvoltage relay to operate under transient overvoltage conditions.	Check voltage regulator base contacts. Check relays as at 3.(b).	Carry out voltage regulator adjustment procedure. Replace defective relay as required.
(b)	Excessive load unbalance.		Refer to item 5.(a).
5. (a) Generator fails to load share.	Improperly adjusted voltage regulator.		Adjust voltage regulator. Difference in regulator settings should be less than 0.2 volts.
(b)	Poor connection of generator ground line in rear of nacelle.	Security of ground line to shunt and of shunt to airframe.	Torque nuts to 110 – 124 pound-inches. Ensure shunt contact surfaces are clean.
6.			Reset (always set generators to “ON” separately). Check battery charge condition.
6. Damage to external power relay, battery relay, battery shunt and associated bus bars.	Reverse polarity of ground power unit (Pre Mod 6/1293).		Replace damaged components. As an insurance against occurrence install Mod 6/1293.
7. No external power input	Open circuit diode – external power control input circuit (Post Mod 6/1293).	Power available at terminal X1 of external power relay.	Replace diode if defective.

Print Date: 2025-09-12

5. Adjustment/Test

A. Voltage Regulator Adjustment

Note The following adjustment procedure must only be done if a new or overhauled voltage regulator is fitted, or if the load sharing is out of limits (between 10 – 20 amps).

Voltage regulators are normally pre-set to 28.5 volts. When operating in extremely high or low temperatures, slight variation in voltage settings as shown in Table 2 — Temperature/Voltage Settings, will assist in preserving battery condition.

Table 2 Temperature/Voltage Settings

Temperature	Voltage Settings
32°F and below	29.0 volts
33°F to 80°F	28.5 volts
81°F and above	28.0 volts

1) Connect generators on-line (refer to Para 5.B – Function Test DC Generation System).

2) Switch off all loads.

3) Set BUS TIE switch to OPEN.

4) Remove plug button from regulation adjustment control access hole.

5) Loosen locknut on adjustment control using a 5/16 inch hexagon nut drive with a hollow shaft.

6) Adjust voltage in accordance with Table 2 — Temperature/Voltage Settings.

7) Select appropriate power lever to IDLE RPM (N_G) and back to idle (N_G) + 15% to ensure voltage is stable (± 0.2 volts).

CAUTION IT IS IMPORTANT THAT THE RELEVANT GENERATOR LOAD IS LESS THAN 0.5 WHEN THE APPROPRIATE POWER LEVER IS SET TO IDLE RPM.

8) Repeat adjustment procedure Para 5.A.4) to Para 5.A.7) for second generator voltage regulator.

9) Parallel generators by setting BUS TIE switch to NORMAL.

10) Ensure appropriate power levers are set to IDLE (N_G) + 15%.

CAUTION IT IS IMPORTANT THAT THE ENGINES ARE MAINTAINED AT THIS SPEED, AND NOT REDUCED, DURING THE FOLLOWING ADJUSTMENT PROCEDURE.

11) Connect as much load as possible.

12) Verify that each generator is load sharing between 10 – 20 amps (= 20 amps).

13) If load sharing is not within limits, individually and alternately adjust (in small increments) the regulation controls by balancing the high load generator (decrease) against the low load generator (increase) until the limits in Para 5.A.12) are achieved.

Note If a generator comes off-line, repeat Para 5.A.6) and Para 5.A.7) and then repeat balancing procedure but in smaller increments.

14) On completion of adjustment, carefully tighten the regulator adjustment locknut, ensuring the setting is not altered.

CAUTION DO NOT OVERTIGHTEN CONTROL LOCKNUT.

15) Recheck load sharing.

16) Disconnect generators (refer to Para 5.B) and shutdown engines (refer to 71-00-00, Power Plant (Mod 6/2012) – General Data).

B. Function Test DC Generation System

1) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

2) Start left engine (refer to 71-00-00, Power Plant (Mod 6/2012) – General Data). Verify no Left Generator Off CAS message is displayed while START switch is held at ON.

3) Release START switch, verify the advisory cyan Left Generator Off advisory CAS message is displayed, and advance power lever to IDLE (N_G) + 15%.

Note After each engine has been running for 90 seconds the advisory CAS message is replaced by a caution CAS message.

Note While either generator is selected off note that the Generators Off indication is shown in the electrical panel.

4) When the engine speed has stabilized, momentarily select the LEFT GENERATOR control switch to RESET and then release to the ON position.

5) Verify no Left Generator Off CAS message is displayed and the Left Generator Amps displays a positive load.

6) If the battery was used to start the left engine and there is any reason to doubt the capability of the battery for starting the right engine, maintain the engine at IDLE + 15% to charge the battery. When the battery charge reading is less than 40 amperes on the Main Battery Amps display, select the LEFT GENERATOR control switch to OFF and verify the amber Left Generator Off caution CAS message is displayed.

7) Start right engine (refer to 71-00-00, Power Plant (Mod 6/2012) – General Data). Verify no Right Generator Off CAS message is displayed while START switch is held at ON.

8) Release start switch, verify the cyan Right Generator Off advisory CAS message is displayed, and advance power lever to IDLE (N_G) + 15%.

9) When the engine speed has stabilized, momentarily select the RIGHT GENERATOR control switch to RESET and then release to the ON position.

10) Verify no Right Generator Off CAS message is illuminated and the Right Generator Amps displays a positive load.

11) Momentarily select LEFT GENERATOR control switch to RESET and then release to the ON position. Verify no Left Generator Off CAS message is illuminated and the Left Generator Amps displays a positive load. Verify the Generators Off is not illuminated.

12) With aircraft loads of approximately 50 amps per bus selected, verify that loads are approximately evenly distributed (within 10 – 20 amps) between the two generators.

13) If external power has been used to start engines, verify the POWER SOURCE switch is set to BATTERY (Pre Mod 6/2314), verify the BATTERY switch is set to ON then set EXTERNAL switch to OFF (Post Mod 6/2314). Verify Main Battery Amps indicates a battery charging condition indicated by a positive reading.

14) Select LEFT GENERATOR control switch to OFF and verify the amber Left Generator Off caution CAS message is displayed and the Right Generator Amps displays as increased reading (i.e., all loads supplied by right generator).

15) Momentarily select LEFT GENERATOR switch to RESET and then release to the ON position. Verify no Left Generator Off CAS message is illuminated and the Left Generator Amps displays a positive load and is balanced with Right Generator Amps.

16) Select RIGHT GENERATOR switch to OFF and verify the amber Right Generator Off caution CAS message is displayed and the Left Generator Amps displays an increased reading (i.e., all loads supplied by right generator).

17) Select BUS TIE switch to OPEN and verify loss of right DC bus. Verify the Bus Tie Open indication is displayed.

18) Reset BUS TIE switch to NORMAL. Verify Right DC Bus is restored and Bus Tie Open indication is not displayed.

19) Select LEFT GENERATOR control switch to OFF and verify the amber Left Generator Off caution CAS message is displayed and the Main Battery Amps displays discharge indicated by a negative reading (i.e. all loads supplied by battery).

20) Shutdown engines (refer to 71-00-00, Power Plant (Mod 6/2012) – General Data).

21) Select MASTER switch and POWER SOURCE switch to OFF (Pre Mod 6/2314); or select EXTERNAL switch to ON (Post Mod 6/2314).

22) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

C. Bench Test Reverse Current Relay

Refer to Figure 5 for details.

Equipment required:

- Direct current generator, 0 – 28 volts 300 amperes
- Voltage regulator
- SPDT switch No. 3
- Generator field rheostat
- SPDT switch No. 5
- SPDT switch No. 6
- 0 – 30 Volt DC voltmeter
- SPST switch No. 8
- SPDT switch No. 9
- 0 – 5 Volt DC voltmeter
- 300 – 0 – 30 ampere DC ammeter
- 300 ampere at 28 volts load bank
- 0 – 100 millivolts DC millivoltmeter
- 24 Volt DC battery

1) Differential Voltage and Reverse Current Test

a) Connect relay.

b) Set all switches to OFF position.

c) Operate generator at a speed between 3000 and 4500 revolutions per minute.

d) With switch (3) in position 1, adjust generator voltage by means of rheostat (4) so that voltage is approximately one volt less than battery voltage.

Note The voltages of battery (14) and generator (1) can be read and readily compared by means of placing switch (6) alternately in positions 1 and 2 and noting reading on voltmeter (7).

e) With switch (5) in position 1 and switch (8) closed, place switch (9) in position 1 and slowly increase generator voltage by means of the rheostat (4). When cutout operates, voltage indicated on voltmeter (10) will go to zero. Differential voltage is indicated just before voltmeter reading goes to zero and should be between 0.35 and 0.65 volts.

CAUTION SWITCH (9) MUST BE OPENED BEFORE NEXT TEST.

f) Measure reverse current opening by slowly decreasing generator voltage by means of rheostat (4) until cutout opens. Reverse current is read on ammeter (11) just before reading goes to zero. Reverse current opening should be between 9 and 25 amperes when generator voltage is at 26 volts. If generator voltage read on voltmeter (7) with switch (6) in position 2 is not 26 volts, use Table 3 — Reverse Current Opening to obtain the corresponding reverse current opening for various voltages.

Table 3 Reverse Current Opening

GENERATOR VOLTAGE (volts)	REVERSE CURRENT (amperes)	
	Min	Max
28	11	27
26	9	25
24	7	23
22	5	21

g) Test for opening of differential relay due to removal of power to “SW” terminal, first repeat procedure Para 5.C.1).c) through Para 5.C.1).e).

h) Decrease generator voltage by means of rheostat (4) until current reading on ammeter (11) has decreased to approximately zero, then open switch (18).

i) Again decrease generator voltage, dropping out contactor, then continue to decrease voltage to lowest value possible.

j) Open switch (5). Differential contacts now should have opened. To check for open contacts, place ohmmeter across terminals “SW” and “APP”, and note resistance reading. Ohmmeter should read approximately 200 ohms. If contacts remained closed, ohmmeter will read zero resistance.

2) Voltage Relay Tests

a) Set all switches to OFF position.

b) Operate generator at speed between 3000 and 4500 rpm.

c) Set switches (3 and 5) in position 1, and switch (6) in position 2. Slowly vary voltage by means of rheostat (4). Closing voltage should be between 20 and 24 volts and is read on voltmeter (7) when relay operates. Opening voltage should be above 18 volts.

d) Connect ohmmeter from “GEN” terminal to “T” test terminal to indicate when voltage relay contacts have closed. Remove name plate and make a connection to terminal “T” by means of 6-32 screw, 1/2–inch or longer, screwed into tapped hole. This hole is located approximately midway between

terminals "APP" and "SW". File point on screw to be sure of a positive connection. After testing, remove screw and replace name plate.

3) Contactor Test

a) Set all switches to OFF position.

Note This test should be made before contactor coil has become heated, and with relay placed such that four mounting holes are flat on table. Position of relay will have some effect on this operating voltage, even though relay may be mounted in any position in aircraft.

b) Operate generator at speed between 3000 and 4500 rpm.

c) Place switch (3) in position 1 switches (5 and 6) in position 2, then vary voltage by means of rheostat (4) to close and open contactor. Closing voltage, read on voltmeter (7), should be between 14 and 15.5 volts. Opening voltage should be 5 volts or below.

4) Millivolt Drop Test

a) Set all switches to OFF position.

b) Operate generator at speed between 4500 and 6000 rpm.

c) Place switch (3) in position 2, and adjust regulator to 28.5 volts.

d) Place switch (5) in position 2, then close load bank switches until current of 300 amperes is read on ammeter (11).

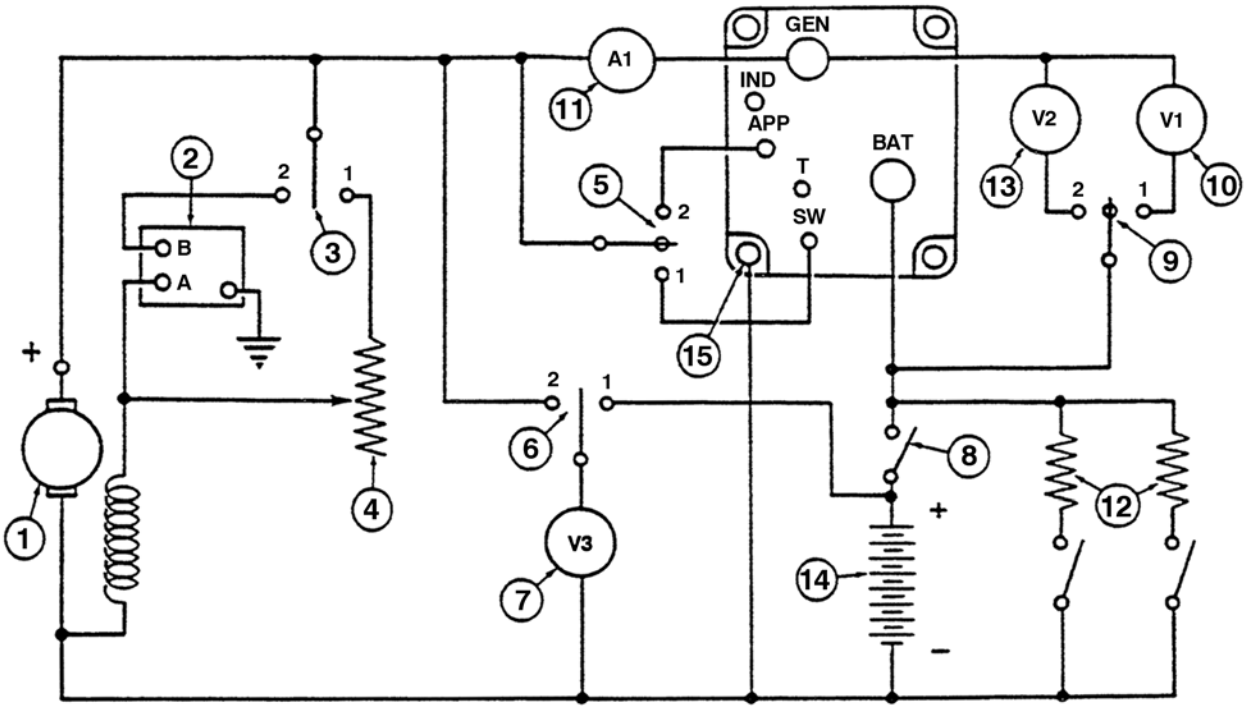
e) Place switch (9) in position 2 and read millivolt drop on millivoltmeter (13). Drop should not exceed 100 millivolts.

CAUTION PLACE SWITCH (9) IN OPEN POSITION BEFORE PERMITTING RELAY TO OPEN. DIFFERENTIAL RELAY CONTACTS SHOULD BE LEFT IN OPEN POSITION TO PREVENT DAMAGE FROM A REVERSE POLARITY GENERATOR.

5) Dielectric Test

a) Using suitable dielectric tester, apply voltage of 1250 volts rms, commercial frequency, for period of one second between following points: each terminal to cover; "BAT" terminal to terminals "IND", "APP" and "SW"; "GEN" terminal to terminals "IND", "APP" and "SW". No breakdown shall occur as a result of this test.

b) Apply 500 volt rms, commercial frequency, for a period of one second between terminals "GEN" and "BAT". No breakdown shall occur.



1. 28V DC 300 AMPERE GENERATOR
2. VOLTAGE REGULATOR
3. SPDT SWITCH
4. GENERATOR FIELD RHEOSTAT
5. SPDT SWITCH
6. SPDT SWITCH
7. 0-30V DC VOLTMETER
8. SPST SWITCH
9. SPDT SWITCH
10. 0-5V DC VOLTMETER
11. 300-0-30 AMPERE DC AMMETER
12. 300 AMP AT 28 VOLTS LOAD BANK
13. 0-100 MILLIVOLTS DC MILLIVOLTMETER
14. 24V DC BATTERY
15. NEGATIVE COVER CONNECTION

Reverse Current Relay Test Circuit
Figure 5

D. Function Test Battery Reserve Polarity Protection Diode (CR109)

Refer to PSM 1-64-2W for electrical schematic.

The following procedure checks the functionality of the battery reserve polarity protection diode (CR109).

Note Diodes that do not pass this test are considered defective and must be replaced.

Equipment Required:

- Digital multimeter, Fluke 87V or similar

1) Aircraft with S.O.O. 6371 (Main Battery installed in Nose Compartment)

a) Ensure aircraft power is OFF and ground power source is disconnected.

Note If S.O.O. 6431 is incorporated, remove upper and lower bulkhead panels, removing the bolts which attach the bulkhead panels to the securing brackets.

b) Disconnect the main battery.

c) Set the MASTER switch to ON and POWER SOURCE switch to BATTERY (Pre Mod 6/2314); or set BATTERY switch to ON (Post Mod 6/2314). This step connects the cathode side of the diode to airframe ground.

d) Gain access to the electrical panel inside the nose compartment at station –6.0.

e) Remove the protective cover to gain access to the main battery relay (K106).

f) Using a digital multimeter in diode test mode, connect the positive test lead to K106 terminal X2 and the negative test lead to airframe ground. This step puts the diode in a forward biased condition.

g) Confirm the forward-bias reading is between 0.5 and 0.8V.

h) Reverse the test leads by connecting the positive test lead to airframe ground and the negative test lead to K106 terminal X2. This step puts the diode in a reverse biased condition.

i) Confirm the reverse bias reading is an open circuit.

j) Disconnect the multimeter and re-install the protective cover at station –6.0.

k) Set the MASTER switch and POWER SOURCE switch to OFF (Pre Mod 6/2314); or set the BATTERY Switch to OFF (Post Mod 6/2314).

l) Reconnect the main battery.

Note If S.O.O. 6431 is incorporated, re-install upper and lower bulkhead panels.

2) Aircraft without S.O.O. 6371 (Main Battery installed underneath floor of rear baggage compartment)

a) Ensure aircraft power is OFF and ground power source is disconnected.

b) Remove the battery compartment floor panel to gain access to the main battery and main battery relay (K106) (refer to 12-00-00, Servicing – General Data, Figure 2 – Servicing Diagram).

c) Disconnect the main battery.

d) Set the MASTER switch to ON and POWER SOURCE switch to BATTERY (Pre Mod 6/2314); or set BATTERY switch to ON (Post Mod 6/2314). This connects the cathode side of the diode to airframe ground.

e) Using a digital multimeter in diode test mode, connect the positive test lead to K106 terminal X2 and the negative test lead to airframe ground. This puts the diode in a forward biased condition.

f) Confirm the forward-bias reading is between 0.5 and 0.8V.

g) Reverse the test leads by connecting the positive test lead to airframe ground and the negative test lead to K106 terminal X2. This puts the diode in a reverse biased condition.

h) Confirm the reverse bias reading is an open circuit.

i) Disconnect the multimeter.

j) Set the MASTER switch and POWER SOURCE switch to OFF (Pre Mod 6/2314); or set the BATTERY Switch to OFF (Post Mod 6/2314).

k) Reconnect the main battery and re-install the floor panel.

E. Function Test Ground Power Source Reserve Polarity Protection Diode (CR108)

Refer to PSM 1-64-2W for electrical schematic.

The following procedure checks the functionality of the battery reserve polarity protection diode (CR108).

Note Diodes that do not pass this test are considered defective and must be replaced.

Equipment Required:

- Digital multimeter, Fluke 87V or similar

1) Aircraft with S.O.O. 6371 (Main Battery installed in Nose Compartment)

   a) Ensure aircraft power is OFF and ground power source is disconnected.

      Note If S.O.O. 6431 is incorporated, remove upper and lower bulkhead panels, removing the bolts which attach the bulkhead panels to the securing brackets.

   b) Disconnect the main battery.

   c) Gain access to the external ground power receptacle inside the forward avionics compartment by removing the HIRF protective covers.

   d) Remove the protective cover next to the external ground power receptacle to gain access to the external power relay (K105).

   e) Using a digital multimeter in diode test mode, connect the positive test lead to the control pin terminal on the external ground power receptacle and the negative test lead to K105 terminal X1. This puts the diode in a forward biased condition.

   f) Confirm the forward-bias reading is between 0.5 and 0.8V.

   g) Reverse the test leads by connecting the positive test lead to K105 terminal X1 and the negative test lead to the control pin terminal on the external ground power receptacle. This puts the diode in a reverse biased condition.

   h) Confirm the reverse bias reading is an open circuit.

   i) Disconnect the multimeter and re-install the protective covers.

   j) Reconnect the main battery.

      Note If S.O.O. 6431 is incorporated, re-install upper and lower bulkhead panels.

2) Aircraft without S.O.O. 6371 (Main Battery installed in AFT Battery Compartment)

   a) Ensure aircraft power is OFF and ground power source is disconnected.

   b) Remove the floor panel to gain access to the main battery and external power relay (K105) (refer to 12-00-00, Servicing – General Data, Figure 2 – Servicing Diagram).

   c) Disconnect the main battery.

   d) Using a digital multimeter in diode test mode, connect the positive test lead to the control pin terminal on the external ground power receptacle and the negative test lead to K105 terminal X1. This puts the diode in a forward biased condition.

   e) Confirm the forward-bias reading is between 0.5 and 0.8V.

   f) Reverse the test leads by connecting the positive test lead to K105 terminal X1 and the negative test lead to the control pin terminal on the external ground power receptacle. This puts the diode in a reverse biased condition.

g) Confirm the reverse bias reading is an open circuit.

h) Disconnect the multimeter.

i) Reconnect the main battery and re-install the floor panel.

F. Power Feeder Cable Inspection at Station 111 Cabin Floor Cut-Out

The following inspection is only applicable to aircraft with S.O.O. 6371 (Forward Relocation of Main Battery and GPU Receptacle).

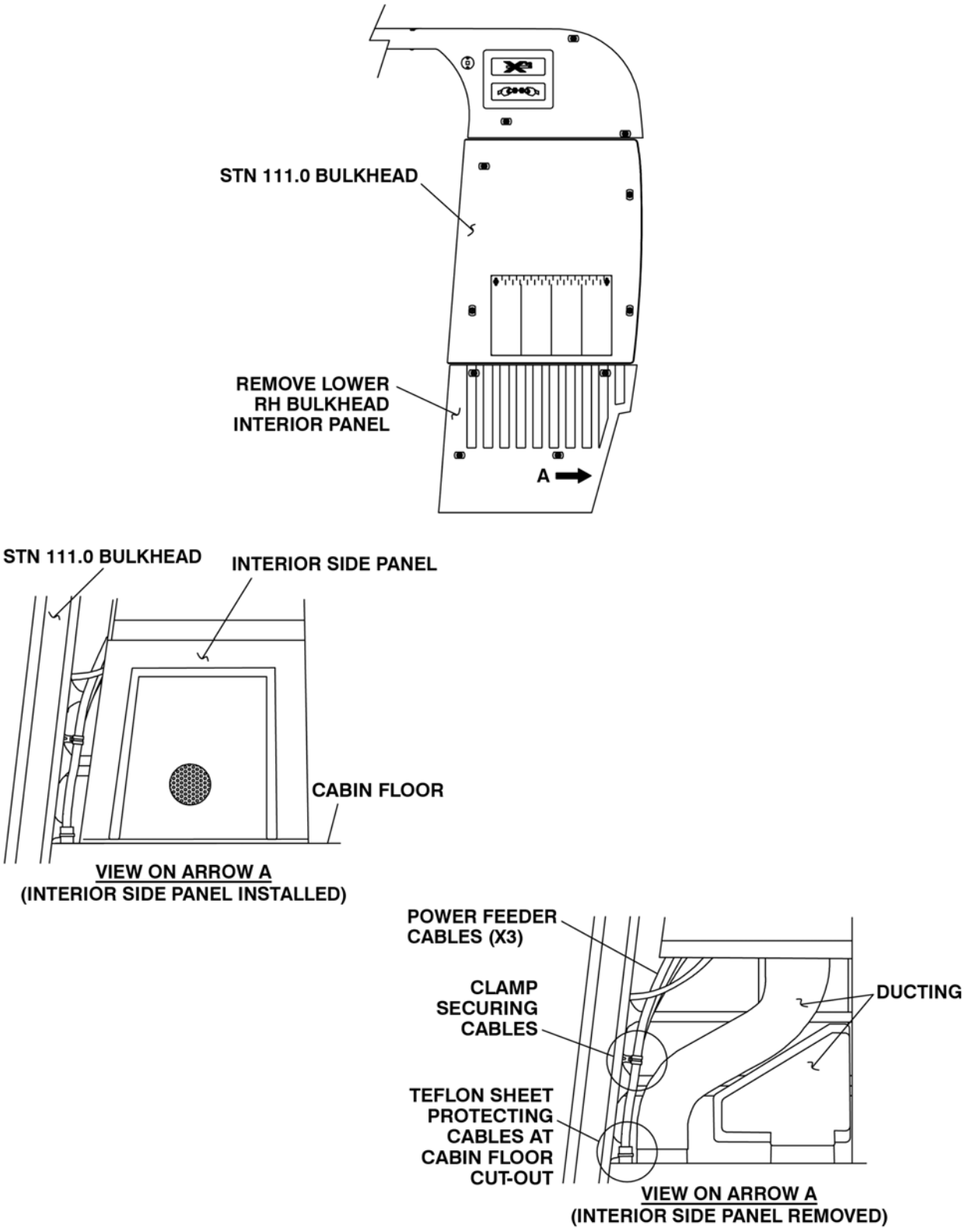
1) Remove the station 111 bottom RH bulkhead interior panel and side panel as shown in Figure 6.

2) Inspect the three power feeder cables near the cabin floor cut-out to ensure there are no signs of mechanical impact, chafing, or overheating.

3) Verify the teflon sheet is securely installed around the cable bundle as shown in Figure 6.

4) Verify the clamp above the cabin floor cut-out is securely installed around the cable bundle as shown in Figure 6.

5) Reinstall the interior panels.



Power Feeder Cable Inspection at Station 111 Cabin Floor Cut-Out
Figure 6

6. Inspection/Check

Refer to Figure 7 for details.

A. Check Installation and Back-to-Back Ratio of Diodes

1) Ensure that electrical power is disconnected, and check for correct polarity of ohmmeter leads on meter to be used for checking.

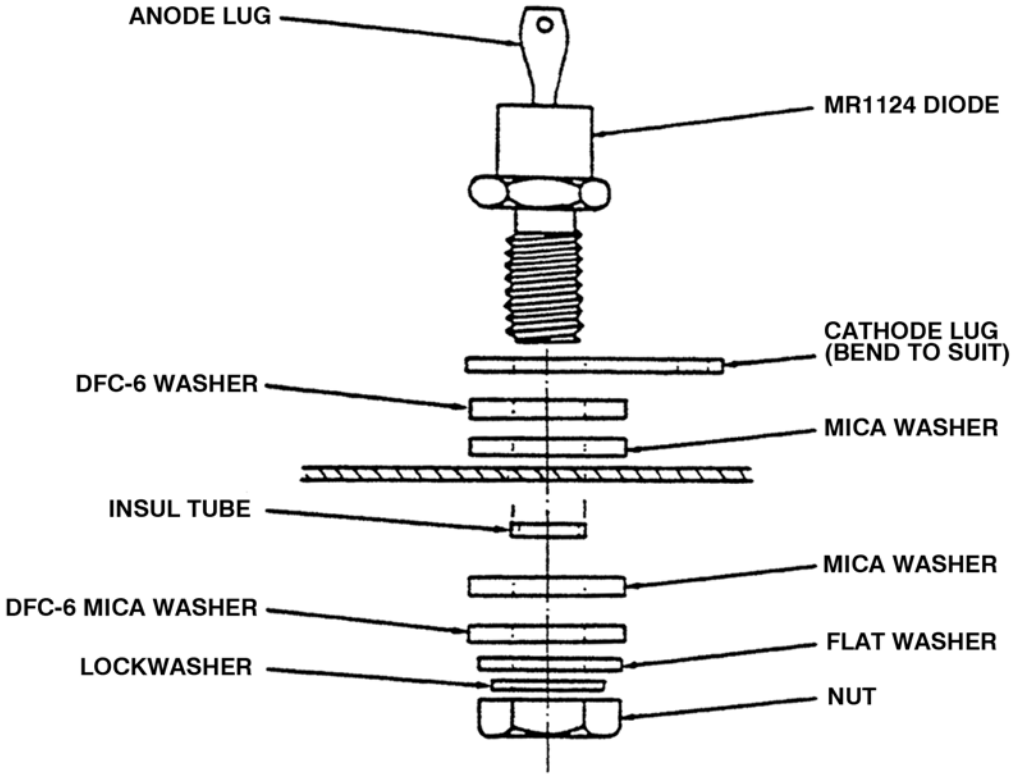
2) Check that blocking diode is in series with load (i.e., anode towards power source).

3) Check that voltage spike suppression diode is connected in parallel with solenoid or relay coil, (i.e., cathode towards power source).

4) When diode is connected with parallel circuit (e.g. soldered across relay coil) apply meter positive lead to anode and negative lead to cathode, and check for low resistance. Then apply negative lead to anode and positive lead to cathode and check that resistance measured is that of the parallel circuit.

5) If possible, disconnect diode from any parallel circuitry, apply meter positive lead to anode and negative lead to cathode of diode, and check for low resistance. Then apply negative lead to anode and positive lead to cathode and check for infinity.

6) When installing MR1124 diodes identified CR5 and CR6 in the main distribution box, ensure that diode mounting holes are properly deburred and install diodes in accordance with Figure 7 and the wrench torque values given in Table 4 — Wrench Torque Values for Terminal Nuts for MR1124 Diodes.



CR5 and CR6 (MR1124) Diode Installation
Figure 7

Table 4 Wrench Torque Values for Terminal Nuts for MR1124 Diodes

STUD SIZE	TORQUE (POUND–INCHES)	
	MINIMUM	MAXIMUM
10/32 inch	12	15
1/4 inch	25	30

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